SHU ZHAO

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Senior Machine Learning Engineer at NVIDIA researching **LLM post-training for reliable, long-horizon agents**, with a focus on reinforcement learning and on-policy distillation. Ph.D. in Computer Science from Penn State. First-author publications at NeurIPS *(Spotlight)*, ICLR, AAAI, IJCAI, and ACM MM *(Oral)*.

SELECTED RESEARCH

LLM Post-Training and Agent Learning

- **R2S** (*arXiv, 2026*): identified a *rollout–scorer gap* in agentic on-policy distillation: a capable teacher assigns high scores to delaying actions that its own rollouts avoid.
  - Developed reward-free teacher repair by mining high-disagreement student states, running teacher takeovers, and fine-tuning the teacher to provide takeover-informed token scores before distillation.
  - Improved success rates by **12.8 percentage points on average** over three agentic-OPD baselines across 12 teacher–student–environment settings in ALFWorld, WebShop, and ScienceWorld. All evaluated distilled students surpassed their teachers; a 2B student achieved 77.9% vs. its 9B teacher's 63.1% on ALFWorld.
- **ExpandSearch** (*arXiv, 2025*): proposed RL-based query expansion with a frozen summarizer to compress retrieved evidence; improved mean exact match by **4.4 points** over the best baseline per model configuration across seven QA benchmarks and four model configurations.
- **ParallelSearch** (*arXiv, 2025*): designed RLVR training for query decomposition and parallel search; improved mean exact match by **2.9 points** over the best baseline per model configuration across seven QA benchmarks. Used **30.4% fewer LLM calls** than Search-R1 on parallelizable subsets.
- **Terminal and coding agents** (*ongoing*): investigating post-training methods for reliable, long-horizon task execution in interactive environments.

Selected Multimodal and Retrieval Research

- **SMILE** (*NeurIPS 2026* ***Spotlight***): proposed visual in-context learning from a VLM's own past errors, without weight updates; improved average task scores by **3.09 points** over a prior visual in-context learning baseline across four benchmarks and four VLMs, with approximately **5× faster inference**.
- **MoRA** (*ICLR 2026*): proposed modality-shared and modality-specific low-rank adaptation for recognition with missing modalities; achieved an average **5.24% relative improvement** over a prior prompt-based baseline across three benchmarks while tuning **0.11% of parameters**.
- **KALAHash** (*AAAI 2025*): designed class-text-guided low-rank adaptation and discrete optimization for low-resource image retrieval; achieved **4× data efficiency**.

OPEN-SOURCE FRAMEWORKS

- **miles-agent**: created a framework for long-horizon agent post-training, integrating miles with NeMo Gym and Harbor for RL, on-policy distillation, and sandboxed terminal/coding rollouts.
- **ferret**: created a verl-based RL framework for search agents, with modular rewards and tools and end-to-end training/evaluation recipes for ParallelSearch, ExpandSearch, and Search-R1.

SKILLS

Post-Training: RL with verifiable rewards, on-policy distillation, reward design, agent evaluation

Agent Systems: multi-turn tool use, search agents, RAG, sandboxed terminal/coding environments

Tools: Python, PyTorch, verl, slime, miles, Harbor, NeMo Gym

Multimodal: VLM adaptation, visual in-context learning, missing-modality robustness, image retrieval

EXPERIENCE

NVIDIA *Santa Clara, CA, USA*

Senior Machine Learning Engineer *Feb. 2026 — Present*

- Lead research on agent post-training (R2S; terminal and coding agents) and multimodal in-context learning (SMILE), from ideation and method design through experiments, paper writing, and presentation.
- Develop and release agent-training frameworks, code, and model checkpoints; build enterprise LLM agents for personal assistance and supply-chain planning.

Machine Learning Engineer Intern *May 2025 — Aug. 2025*

- Initiated and led RL research that developed into ParallelSearch and ExpandSearch, exploring parallel query decomposition and query expansion with evidence compression.
- Designed training objectives and conducted model training and evaluation experiments.
- Built a deep-research agent over an internal bug-report knowledge base to retrieve related reports and draft diagnostic summaries.

Baidu Research *Beijing, China*

Research Intern *Apr. 2022 — Jul. 2022*

- Investigated adaptation of web-scale vision-language models for enterprise image retrieval.
- Insights from this work later informed KALAHash (AAAI 2025), a method for low-resource image retrieval.

SELECTED PUBLICATIONS

- **Shu Zhao**, Tan Yu. "Capable Teacher, Myopic Scorer: Reward-Free Teacher Repair for Agentic On-Policy Distillation." arXiv, 2026.
- **Shu Zhao**, Tan Yu. "The Value of Being Wrong: Self-Mined Visual In-Context Learning from Errors." **NeurIPS 2026** ***(Spotlight)***.
- **Shu Zhao**, Nilesh Ahuja, Tan Yu, Tianyi Shen, Vijaykrishnan Narayanan. "MoRA: Missing Modality Low-Rank Adaptation for Visual Recognition." **ICLR 2026**.
- **Shu Zhao**, Tan Yu, Anbang Xu. "Beyond the Limitation of a Single Query: Train your LLM for Query Expansion with Reinforcement Learning." arXiv, 2025.
- **Shu Zhao**, Tan Yu, Anbang Xu, Japinder Singh, Aaditya Shukla, Rama Akkiraju. "ParallelSearch: Train your LLMs to Decompose Query and Search Sub-queries in Parallel with Reinforcement Learning." arXiv, 2025.
- Xiaoshuai Hao, Lingdong Kong, Rong Yin, Pengwei Wang, Jing Zhang, Yunfeng Diao, **Shu Zhao**. "SafeMap: Robust HD Map Construction from Incomplete Observations." **ICML 2025**.
- Xiaohan Zou, Wenchao Ma, **Shu Zhao**. "Learning Conditional Space-Time Prompt Distributions for Video Class-Incremental Learning." **CVPR 2025** ***(Highlight)***.
- **Shu Zhao**, Tan Yu, Xiaoshuai Hao, Wenchao Ma, Vijaykrishnan Narayanan. "KALAHash: Knowledge-Anchored Low-Resource Adaptation for Deep Hashing." **AAAI 2025**.
- **Shu Zhao**, Dayan Wu, Yucan Zhou, Bo Li, Weiping Wang. "Rescuing Deep Hashing from Dead Bits Problem." **IJCAI 2021**.
- **Shu Zhao**, Dayan Wu, Wanqian Zhang, Yu Zhou, Bo Li, Weiping Wang. "Asymmetric Deep Hashing for Efficient Hash Code Compression." **ACM MM 2020** ***(Oral)***.

Full publication list on Google Scholar.

EDUCATION

Ph.D. in Computer Science, The Pennsylvania State University *Aug. 2022 — Dec. 2025*

M.E. in Computer Technology, University of Chinese Academy of Sciences *Sep. 2018 — Jun. 2021*

B.E. in Information Security, Anhui University *Aug. 2013 — Jun. 2017*